



SUPPORT REQUEST

VIRTUAL CONTROL
BROADCOM SUPPORT SUBMISSION

NSX 9.0.1 Orphan vDS Cleanup Cannot Complete

Three different NSX 9.0.1 API cleanup paths (action=clear ownership, per-host TN PUT, TNC DELETE) all fail with three different undocumented error codes — request for force-clear / vLCM-aware detach procedure.

NSX 9.0.1

VCF 9.0.1

ERROR 8012

ERROR 9560

ERROR 26185

SR Draft

Ready for case submission

Broadcom Support Request — NSX 9.0.1 Orphan vDS Cleanup

Submission Information

Field	Value
Product	VMware NSX 9.0.1 (deployed by VMware Cloud Foundation 9.0.1)
Component	Manager API — managed-objects/switching/nsx-ownership , host-transport-nodes , transport-node-collections
Severity (proposed)	Severity 3 — operational impact (orphan record blocks routine cleanup, does not break running traffic)
Environment	Lab / non-production; VCF 9.0.1 nested on VMware Workstation; 4 ESXi hosts; vSAN OSA
Submission ready date	2026-05-16
Submitter	Virtual Control LLC
Internal reference	VCL-SR-2026-0516-NSX-ORPHAN-VDS

1. Executive Summary

After a vSphere Distributed Switch was replaced in this cluster (old vDS UUID [ORPHAN-UUID] removed, new vDS vcenter-c101-vds01 with UUID [LIVE-UUID] created), NSX 9.0.1 records still reference the old UUID at three layers:

1. **NSX ownership table** for Compute Manager [CM-UUID] includes the orphan vDS UUID.
2. **NSX Transport Node Profile (TNP)** for the cluster had host_switch_id pointing at the orphan UUID. Fixed in-place by PUT (succeeded).
3. **Per-host Transport Nodes (TNs)** for esxi01–esxi04 still carry the orphan host_switch_id in their host_switch_spec . Cannot be updated.

Three documented NSX API cleanup paths have been attempted; **each fails with a different error code that has no public Broadcom KB**:

Path	API	Error
Clear NSX ownership of orphan vDS UUID	PUT /api/v1/managed-objects/switching/nsx-ownership/<cm>?action=clear with mo_id=<orphan-uuid>	HTTP 400 — error_code 8012 "could not be found"
Update per-host TN host_switch_id from orphan to live	PUT /policy/api/v1/infra/sites/default/enforcement-points/default/host-transport-nodes/<tn-id>	HTTP 400 — error_code 9560 "TZ cannot be relocated between HostSwitches in a single transaction"
Detach TNP from cluster (force re-prep)	DELETE /api/v1/transport-node-collection/s/<tnc-id>	HTTP 400 — error_code 26185 "Transport Node Profile cannot be detached from vLCM enabled cluster"

The `action` parameter on the ownership endpoint is enum-locked to `[set, clear]` (verified — `force_clear` and `clear_local` rejected with error 255). No `force=true` parameter (rejected with error 268 "unknown parameter").

Request: Documented procedure (or supported `force=true` / `local-only=true` flag, or NSX Manager CLI / `nsxapi` command) to remove the orphan UUID from NSX's ownership table without invoking the broken `clear` action that requires vCenter to acknowledge an object it does not have.

2. Environment

Sanitization note: UUIDs and the Compute Manager ID below are real lab values; passwords have been redacted. Lab is non-production. Full UUIDs preserved because they are needed to reproduce the issue against the actual NSX manager state in any escalation snapshot Broadcom may take.

Component	Detail
VCF Version	9.0.1.0
NSX Manager build	9.0.1 (verify exact build at log capture; <code>GET /api/v1/node</code> returns <code>node_version</code>)
NSX Manager IP	192.168.1.71 (single-node lab; not 3-node cluster)
vCenter	vcenter.lab.local (192.168.1.69), build 24957454 (vSphere 9.0.1)
Compute Manager ID in NSX	259b8edb-3b54-496f-bfb4-c85c145cd9bb
Cluster ID (vCenter MoRef)	domain-c10
Cluster name	vcenter-cl01
Live vDS name	vcenter-cl01-vds01
Live vDS UUID	50 12 35 19 fa f5 be 0a-71 8d ab e0 b5 e6 b9 cf
Orphan vDS UUID	50 37 13 3f 96 e4 3e 93-1d f0 3b 01 d4 76 6a f8
TNP ID	b7ac9c27-4620-4a74-bd2d-0e111c56493c (name vcenter-cl01-tnp-01)
TNC ID	12a28dea-427a-4502-9751-4f3800987f1b (name vcenter-cl01-tnc)
Cluster lifecycleManaged flag (NSX-side <code>compute_collection.origin_properties</code>)	true
Cluster vLCM image (vCenter-side <code>ClusterConfigInfoEx.DesiredImageSpec</code>)	null / unset
ESXi hosts	esxi01.lab.local ... esxi04.lab.local (4 hosts, all healthy)
NSX Transport Zone IDs (active)	vcenter-cl01-tz-overlay01 , vcenter-cl01-tz-vlan01 , plus auto-generated 9249d111-... (post-TNP-PUT) and orphan 54a8a26a-... (pre-PUT)
NSX TZ realized ID that error 9560 references	3fbf2d31-12a9-4781-8cd3-95b4bf472824
Segments deployed	0

Component	Detail
Edges / T0 / T1 deployed	0

3. Problem Statement

3.1 How the orphan state was created

The cluster's original vDS was replaced (the exact change-control event is not in this lab's records — the orphan UUID `50 37 13 3f...` is no longer in `Get-VDSwitch` output and produces `404 / View not found on direct Get-View -Id "DistributedVirtualPortgroup-dvportgroup-6025"`). The new vDS `vcenter-cl01-vds01` (UUID `50 12 35 19...`) has been operational since. NSX was prepared on the cluster against the original vDS and **was not unprep'd before the vDS replacement**.

3.2 Symptoms observed

1. vCenter alarm: *"The vSphere Distributed Switch corresponding to the proxy switches `50 37 13 3f 96 e4 3e 93-1d f0 3b 01 d4 76 6a f8` does not exist."* on each of esxi01–esxi04.
2. NSX UI / API "Resolve" attempt against the orphan triggers `error_code 8012`.
3. Per-host TN realization shows `host_switch_id = 50 37 13 3f...` even after TNP-level PUT corrected the TNP. TNs are at `_revision: 0`, never reconciled after the original prep.
4. Until 2026-05-16, four management VMs (vcenter, sddc-manager, vcf-ops, fleet) also retained NIC bindings to stale DVPg MoRef keys on the orphan UUID. These were migrated to live-vDS DVPgs today using power-off → reconfigure → power-on (online NIC swap failed with "object or item referred to could not be found" because vCenter cannot release a port on a switch UUID that has no inventory entry — separate but related issue). **All VMs are now confirmed on the live vDS**. The NSX residual is the remaining state.

3.3 What's NOT broken

- All four ESXi hosts: healthy, NSX TN realization state = `success`, TEP vmkernel interfaces (vmk10, vmk11) up on 192.168.13.0/24
- NSX manager cluster: `STABLE`
- TNP after PUT: `_revision: 1`, `host_switch_id` = live vDS UUID
- vCenter alarm referenced above: **cleared** after TNP PUT (alarm reads from TNP, not TN state)
- VCFA / overlay workload deployment: not impacted by the residual TN state

The orphan record is **internal NSX state** with no production-traffic impact, but it blocks any operator action that triggers ownership cleanup workflows.

4. Reproduction — Exact API Calls and Responses

4.1 Path A — Clear NSX ownership (error_code 8012)

Request:

```
PUT /api/v1/managed-objects/switching/nsx-ownership/259b8edb-3b54-496f-bfb4-c85c145cd9bb?action=clear
Content-Type: application/json
Authorization: Basic <admin:redacted>

{"mo_id":"50 37 13 3f 96 e4 3e 93-1d f0 3b 01 d4 76 6a f8"}
```

Response (HTTP 400):

```
{
  "details": "The requested object : 50 37 13 3f 96 e4 3e 93-1d f0 3b 01 d4 76 6a f8 could not be found.
Object identifiers are case sensitive.",
  "httpStatus": "BAD_REQUEST",
  "error_code": 8012,
  "module_name": "NsxSwitching service",
  "error_message": "Unable to clear NSX ownership from MO 259b8edb-3b54-496f-bfb4-c85c145cd9bb/50 37 13 3f
96 e4 3e 93-1d f0 3b 01 d4 76 6a f8."
}
```

Variants attempted (all rejected):

Variant	Result
?action=clear&force=true	HTTP 400 — error_code 268 : "Request has unknown parameters: force."
?action=force_clear	HTTP 400 — error_code 255 : "value force_clear of property action is not one of the allowed values [set, clear]"
?action=clear_local	HTTP 400 — error_code 255 : same
DELETE /api/v1/managed-objects/switching/nsx-ownership/<cm>	HTTP 405 Method Not Allowed
GET /api/v1/managed-objects/switching/nsx-ownership/<cm>	HTTP 405 Method Not Allowed (so we cannot list current ownership records to verify the table)
?action=set with the LIVE vDS UUID as mo_id	HTTP 200 (succeeded — but subsequent ?action=clear with the orphan UUID still returns 8012)

4.2 Path B — Update per-host TN host_switch_id (error_code 9560)

Request (one host, esxi01 TN 7a93666c-fbe2-4d7f-ad94-0c5525ac331f):

```
PUT /policy/api/v1/infra/sites/default/enforcement-points/default/host-transport-nodes/7a93666c-fbe2-4d7f-ad94-0c5525ac331f
Content-Type: application/json

{
  "host_switch_spec": {
    "host_switches": [{
      "host_switch_name": "vcenter-cl01-vds01",
      "host_switch_id": "50 12 35 19 fa f5 be 0a-71 8d ab e0 b5 e6 b9 cf",
      "host_switch_type": "VDS",
      "host_switch_mode": "STANDARD",
      ...rest of payload preserved from prior GET; only host_switch_id and host_switch_name changed...
    }],
    "resource_type": "StandardHostSwitchSpec"
  },
  "_revision": 0
}
```

Response (HTTP 400):

```
{
  "httpStatus": "BAD_REQUEST",
  "error_code": 9560,
  "module_name": "NsxSwitching service",
  "error_message": "Transport Zone TransportZone/3fbf2d31-12a9-4781-8cd3-95b4bf472824 cannot be relocated
from HostSwitch 50 37 13 3f 96 e4 3e 93-1d f0 3b 01 d4 76 6a f8 to HostSwitch 50 12 35 19 fa f5 be 0a-71
```

```
8d ab e0 b5 e6 b9 cf in a single transaction. Delete HostSwitch from current Transport Zone to add later to another Transport Zone."
}
```

This error reproduces on all four TNs identically.

4.3 Path C — Detach TNP from cluster (error_code 26185)

Request:

```
DELETE /api/v1/transport-node-collections/12a28dea-427a-4502-9751-4f3800987f1b
```

Response (HTTP 400):

```
{
  "httpStatus": "BAD_REQUEST",
  "error_code": 26185,
  "module_name": "service-installation-fabric",
  "error_message": "Transport Node Profile cannot be detached from vLCM enabled cluster."
}
```

The gating flag is `origin_properties.lifecycleManaged = true` on the NSX-side `GET /api/v1/fabric/compute-collections/259b8edb-3b54-496f-bfb4-c85c145cd9bb:domain-c10`. **vCenter's `ClusterConfigInfoEx.DesiredImageSpec` is null** (no vLCM image set), so the NSX-side flag appears to be coming from VCF's cluster registration, not from vSphere image lifecycle.

4.4 PowerCLI / vCenter NIC reconfigure on running VM bound to orphan DVPG

While not the primary subject of this SR, the related vCenter-side symptom: `Set-NetworkAdapter -PortGroup pg-mgmt` on a running VM whose current backing references a stale `PortgroupKey` (e.g., `dvportgroup-6025`) returns success at the cmdlet layer but produces an `Error` task in vCenter's task log with message `"The object or item referred to could not be found."` (`ManagedObjectNotFound`). Workaround that works reliably: power off → `ReconfigVM_Task` → power on. KB 418461 describes a similar pattern for the standard-vSwitch → DVS case but does not cover the ghost-DVPG case observed here.

5. What's Been Tried Locally

Attempt	Outcome
PUT TNP <code>b7ac9c27-...</code> with corrected <code>host_switch_id</code>	Succeeded. <code>_revision: 0 → 1</code> . vCenter alarm cleared. NSX automatically regenerated <code>portgroup_transport_zone_id</code> from <code>54a8a26a-...</code> to fresh <code>9249d111-...</code> .
Migrate 4 mgmt VMs (vcenter, sddc-manager, vcf-ops, fleet) off stale DVPG keys to live-vDS DVPGs	Succeeded (power-off → reconfigure → power-on procedure, today 2026-05-16). All 6 VMs on cluster now confirmed bound to live vDS UUID.
Path A — <code>?action=clear</code> on orphan UUID	Failed — 8012 (object not found in vCenter)
Path A variants (<code>force=true</code> , <code>force_clear</code> , <code>clear_local</code> , <code>DELETE</code>)	All rejected (255 / 268 / 405)
Path A — <code>?action=set</code> on live UUID, then re-try <code>clear</code> on orphan	<code>set</code> succeeded with HTTP 200; subsequent <code>clear</code> on orphan still 8012. The <code>set</code> write did not displace the orphan entry.

Attempt	Outcome
Path B — per-host TN PUT to correct <code>host_switch_id</code>	Failed on all 4 hosts — 9560 (TZ relocation guard)
Path B — re-tried after VM migration cleared orphan VM references	Still 9560. Underlying TZ-to-host_switch binding is independent of VM presence.
Path C — TNC DELETE	Failed — 26185 (vLCM-enabled cluster)
Path C — re-tried after VM migration	Still 26185. The <code>lifecycleManaged</code> flag is independent of VM membership.
Search Broadcom KB for <code>error_code 8012</code> , 9560, 26185, <code>nsx-ownership clear</code> , <code>transport node profile vLCM detach</code> , <code>Transport Zone cannot be relocated</code>	No public KB matches for any of the three error codes in the context observed. Related but non-applicable KBs: 387942 (orphan host entries), 387148 (orphan portgroups), 379010 (DVS out-of-sync), 370575 (NVDS→VDS stale switch), 298537 (orphan NSX-T objects). None describe the post-vDS-replacement state captured here.

6. Specific Asks for Broadcom

Numbered for traceability when discussing with engineering:

1. **For error_code 8012 (clear NSX ownership) when the referenced vCenter MO does not exist** — is there: - a supported `force=true` / `local=true` flag I missed? - an NSX Manager CLI command (e.g., `clear database`, `clear vds-binding`) that removes the ownership record without requiring vCenter to acknowledge release? - a documented procedure to remove the orphan record from the NSX manager backend (Postgres / Corfu)?
2. **For error_code 9560 (TZ relocation single-transaction guard)** — is the two-transaction workaround documented anywhere? Specifically: - Step 1: detach TZ from current `host_switch_id` (what API call?) - Step 2: attach TZ to new `host_switch_id`

Or is the only supported path to detach the TNC (which itself is blocked — see ask #3)?

3. **For error_code 26185 (TNP cannot be detached from vLCM-enabled cluster)** — what is the supported workflow to: - either disable the `lifecycleManaged=true` flag on NSX's compute-collection record, or - perform the TNP detach through the lifecycle-aware path (which UI / API endpoint)?

The cluster has no actual vLCM image set in vCenter, so the lifecycle gate appears to be coming from NSX-side state rather than vCenter-side state.

4. **For the general scenario** — is there a documented runbook for "NSX was prepared against a vDS; the vDS was replaced (or its UUID changed) without unprep'ing NSX first; how do I recover NSX records to point at the new vDS"? This appears to be a common-enough scenario that we'd expect a KB or TechDoc, but our searches have not surfaced one for NSX 9.0.x.
5. **For PowerCLI Set-NetworkAdapter silent failure** — is the swallowed `ReconfigVM_Task` error on cached-task return a known issue? It bit us today because the cmdlet returned success with no warning while the vCenter task had errored. We worked around it by going to direct `Get-View / ReconfigVM_Task` for visibility, but other operators might not.

7. Suggested Workaround (if Broadcom recovery path is not available)

If no supported recovery path exists, we propose to:

1. Schedule a maintenance window with 0 NSX segments active (current state).
2. Disable NSX on the cluster via VCF Operations / SDDC Manager UI workflow (the supported lifecycle-aware path), accept the full un-prep + re-prep cycle of ~30 min per host.
3. After re-prep, the TNs will inherit fresh `host_switch_id` from the corrected TNP; ownership table re-syncs against current vDS.
4. Document this in our internal runbook as the "nuclear" path.

Confirmation that this is the only supported path would unblock that plan.

8. Attached Diagnostics (separately)

The following items will be attached to the case as separate files:

Attachment	Contents
<code>nsx-support-bundle-2026-05-16.tar.gz</code>	NSX Manager support bundle generated after all three failures reproduced; captures Postgres/Corfu state and manager.log around the failed API calls
<code>nsx-error-traces.txt</code>	Verbatim HTTP request/response pairs for all three error codes (Paths A, B, C) plus all attempted variants
<code>vcenter-state-2026-05-16.json</code>	Get-VDSwitch + Get-VDPortgroup output showing only the live vDS in inventory; Get-View on orphan DVPG keys returns 404
<code>tnp-tnc-payloads.json</code>	Full GET response payloads for the TNP and TNC, before and after the fix attempts
<code>compute-collection-origin.json</code>	<code>GET /api/v1/fabric/compute-collections/<cc></code> showing the <code>lifecycleManaged=true</code> flag

9. Internal Document Linkage

This SR is grounded in three internal Virtual Control LLC documents that contain the full reproduction story:

- `VCF-Mgmt-Plane-vDS-Migration-Plan-2026-05-16.md` — the migration plan and execution log that demonstrated all four management VMs were on the orphan vDS, and the procedure that moved them off
- `NSX-Trust-CM-TNP-Repair-Plan-2026-05-15.md` — the earlier diagnostic work that uncovered each error code in turn
- `VCF-Undocumented-Issues-Reference.md` Section 12 — formal write-up of the three error patterns as undocumented issues

10. Document Information

Field	Value
Document ID	VCL-SR-2026-0516-NSX-ORPHAN-VDS
Version	1.0
Date	May 16, 2026

Field	Value
Author	Virtual Control LLC
Status	Ready for submission via Broadcom Support Portal
Recommended product / module	VMware NSX → Manager → API
Recommended initial severity	Severity 3 (operational, non-traffic-impacting)

End of submission.